



The Intersection of AI and Consumer Protection

By Jeannie Paterson, Melbourne Law School

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I'm Professor Jenny Patterson from the University of Melbourne in Australia at the moment, I'm on the south coast. And you can probably hear some bellbirds in the background. I'm going to speak today about AI and consumer protection. And I'm going to focus on predictive analytics in targeted advertising. So, in a minute, I'll share my screen. This seminar is going to focus on the uses of data in the processes of selling to consumers. With a combination of big data and AI applications in digital marketing, marketing provides firms with new opportunities for reaching consumers and for influencing consumers' decision making. Firms are able to individualize their pitch to consumers through targeted advertising and through differential pricing, and to vary their advertising and pricing strategies in a very short period of time, in response to granular information about consumer responses. Many consumers do not know this happens or what it involves, and yet are using personal data that fuels precisely these processes. Perhaps this is not a problem. But I think consumers should know what it's happening, and should have a real choice over this process, as well as some strong protections in place for when consumers don't or can't make the decisions that best protect their own interests. This is a subset of the field of research and scholarly consideration that involves surveillance capitalism. You can read more about that in Shoshana Zuboff's work and also by other scholars.

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So what we're gonna look at today are these things. We're gonna look at data collection, we're gonna look at algorithmic advertising, we're gonna ask what the consumer harm is in these processes, and then we're gonna think about some legal responses. Let's start with data collection. Consumers' online and digital interactions provide considerable amounts of personal data that may be collected and sold to inform marketing strategies of goods, media, and even political parties. Traders typically now collect large amounts of data about the consumers they deal with. This may be done through innocuous methods, such as loyalty cards, or memberships. Large corporate retailers collect information through their websites. Data may also be collected through consumers' casual online interactions, including browsing, use of social media, and internet transactions. Retailers and digital platforms can collect large amounts of data about consumers because they have a variety of processes for doing that, that aren't always visible. They can use cookies, web beacons, device or browser fingerprint, even facial recognition and certainly tracking of mobile devices and audio beacons. The data collected from these numerous seemingly discrete interactions may be recombined through sophisticated data analytics to create a fuller picture of consumers' online behavior interest preferences, a process sometimes referred to as consumer profiling. These profiles are then used for a variety of purposes, including to drive platform and product innovation, and for targeted advertising and differential pricing.

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We know that social media tracks and records many of our interactions, but let's not be fooled into thinking that it's just the digital platforms. A forthcoming report by my colleague, Chris Kernaghan, and others identify the scope and scale of online advertising and tracker activity on the landing pages of five Australian retailers sites,

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supermarkets and popular shopping venues. We found multiple advertising networks, trackers, and profiling scripts present on all five retailers surveyed. The report observed that in a retail context in which the consumer is not receiving a product or service for free, but is paying for it, such data collection needs to be further justified to ensure the market remains fair. We know we're the data subject on digital platforms but we're also the data subject of supermarkets, department stores, and other service providers. The lack of transparency in the power asymmetry that arises from the information gathering and balance on retail and digital platform sites are apparent.

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The question is what do we do about this? But before we ask that question, let's delve a little bit deeper into the data collection processes. Two of the main forms of data collection are cookies and pixels. A cookie is a small plain text file sent by a website, such as Amazon, to a browser, such as Chrome. The file is then stored in the user's computer and it's sent back to the website server whenever they return. In this way, cookies can be used to track the user's navigation of website down to the time spent on each web page. This obviously provides valuable information for marketing purposes. Digital platforms such as Facebook make use of third-party cookies. These are cookies that are set by platforms on various websites. They represent the majority of cookies added by websites. This data collection mechanism is enabled by plugins such as the Facebook, Twitter and Pinterest 'like' and 'share' buttons that you see on many websites. In this way data can be collected about users by these digital platforms even if the users or consumers aren't members of those platforms.

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Recently, blocking third party cookies has become easier, and some new solutions have been put in place by these firms and platforms to allow them to continue to collect user data. One such solution is called a web beacon. Facebook calls it the Facebook Pixel. It is an often transparent image which is placed on a website to collect data and track conversion from ads, how to turn off and third party cookies. Cookies and pixels are used to record information about specific users. For example, cookies will store information to identify you as a user, and your operating system and the browser you prefer. This type of information will be collected and linked to this identifier includes the products you click on, add to your cart, buy, or even the time you dwell on particular pages. Since such information is not stored on your device, it stays rather encoded and collected by the platform or firm, it is difficult to determine the nature and extent of data being collected. We know however, that the data is bundled together and then repurposed for advertising and other uses. The data that's collected from consumers' shopping interactions is indeed aggregated and repurposed to inform the marketing strategies for goods, media, and political parties. The data goes deeper than straightforward identifying information such as your name or birthdate.

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It also includes information about where consumers shop, work and socialize, who they interact with, and how long and how often they visit particular web pages. Indeed, from the advertising perspective, intimate information about the name and age of the consumer is less important than information about their social network habits and proclivities. From this perspective, algorithmic advertising is less interested in individuals, then what Amore and Yule term 'data derivatives', namely the algorithmic constructions or abstractions of individual made out of fragments of correlated variables identified from data matches between their attributes collected on websites. Increasingly, the combination of information drawn from hubs to data and predictive analytics processed through these algorithms will enable firms to personalize advertising, offer differential pricing and to provide a variety of tools and/or sending the best product according to consumers needs or wants. Currently, the techniques differ in

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scope as between users and are by no means universal. However, they seem likely to continue and grow in sophistication and reach. Two prominent examples in this field are personalized advertising and differential pricing. Let's dig a little bit more into algorithmic advertising. We're all familiar with the ads that follow you around the internet. This occurs through cookies. For example, when a consumer is viewing a product online, causing them to download cookies onto their computer. That then loads the advertisement that they've been looking at onto other websites that they have subsequently visited. Behavioral advertising is advertising based on predictions about the interests of certain kinds of consumers. The data collected from consumers' online interactions, as I've noted, creates a digital persona. Advertisers use predictive analytics with varying degrees of sophistication to determine the kinds of ads that are likely to appeal to those consumers on the basis of their online personas, and their exhibited behavior. So, famously, Target sending ads about baby products to a young woman who was pregnant before her family even knew she was pregnant. And it did this on the basis of a prediction from her online behavior. The form of the advertisements may even change between different consumers to correlate with their particular preferences.

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It's been found that about 78% of consumers prefer personalized advertising as opposed to non-personalized advertising. However, personalized advertising causes consumer discomfort when there is a low level of transparency as to what data is collected, despite high transparency in the collector of data that violates people's reasonable expectation about what's appropriate to collect and what can be used with it. Another form of algorithmic advertising is targeted advertising. Behavioral advertising sends ads to consumers based on their predicted likelihood of interest. Target advertising is advertising where a firm taking out the ad chooses the demographic. In other words, the firm who's advertising picks who they'd like the advertisement to go to. Digital platforms have large amount of access to large amounts of consumer data and are able to slice and dice that data in different ways.

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Therefore offering all sorts of different target audience segments, for example, based on demographics, prior purchasing activity, interests, behaviors and location. Subtle characteristics may also be detected, for example, categories offered by some digital platforms include people who oppose immigration, have followed politics, or interested in vaccine controversies or climate change denial. A data broker states that it divides Australian households into different consumer segments including affluent adventurers; countryside elite; suburban, thrift and prosperous families. It also has categories that are much more nuanced and focused and potentially open to manipulation such as diabetes interests, cholesterol focus, financially challenged, and urban scramble. Facebook and Snapchat further offer look-alike audiences, which allow firms advertising their wares to target nuances, new audiences based on demographic features of their current customers. This made possible the infamous redlining episodes in the United States where certain kinds of consumers, mainly African American consumers, were not shown ads for rental properties. Finally, differential pricing. Differential pricing is third degree price discrimination. In other words, changing the pricing for individuals according to their predicted willingness and capacity to pay for products based on information collected about them, and consumers who were in similar situations. This kind of development may mean that well-off consumers are charged more for certain products and services. For example, there was some suggestion that well-off consumers were charged more for airline flights and hotels on certain flight and hotel bundling sites. In the same vein, differential pricing can benefit consumers who are not financially strong, such as students, retirees and those from lower socio-economic demographics who may be given discounts for certain high demand products. Differential pricing may also not be so so favorable

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towards the poor. Conceivably, firms may use profiling to charge higher prices to low income consumers, either to discourage them as customers, or as a firm to offer some form of insurance against credit and performance risk perceived to attach to specific groups of customers. Which leads to the next heading - is there any harm in personalized advertising? Further research is needed on how consumers understand online shopping and advertising, and the ways in which online advertising and individual targeted advertising may influence their choices. However, the surveys that we've done in this field show great degree of uncertainty by consumers about whether such practices occur and how they should respond to them. There is potential for a variety of harms arising from targeted advertising, and differential pricing. These include marketing inefficiencies, decreased likelihood of competition, reduced trust in data driven technologies. They also include risks of manipulation and a loss of autonomy. All marketing has the aim in some way of influencing consumers purchasing decisions. Bricks and mortar stores aim to influence consumer choice through their advertising, and by the very way in which the stores display their products, along with the ambience they create inside the store. However, the online environment differs significantly from bricks and mortar shopping. The use of digital technology and marketing to consumers allows firms to go well beyond what would have been possible in the past. As we've seen, firms have the opportunity to target advertising to narrowly defined consumer profiles, with different consumers seeing different material and being offered different prices. By removing alternative options from consumers' side, the capacity for choice is reduced. Consumers may be unaware of an advertisement they see online is only seen by them or people thought to be like them, and that other consumers are seeing different options. This means consumers are making decisions from a position of less than full information, undermining their preconditions for the exercise of autonomy. Targeted advertising may also be very effective in influencing consumer preferences in subtle ways that are not perceived or realized by the consumer. And they do this in the way in which the options available to consumers are presented. And also by being designed to trigger pre-existing sensitivities or unconscious biases identified through data-driven profiles. In this way, online advertising may nudge or even shove consumers towards particular products.

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For example, food or cosmetics may be advertised at particular times of day when consumers are predicted to be feeling tired, stressed or lacking confidence. Targeted advertising also has the potential to entrench existing inequalities in society. It may do this by allowing firms to discriminate between different consumer profiles. This creates risks for vulnerable and marginalized consumers who may be further restricted in their purchasing choices and subject to unfavorable pricing through these methods. For example, a targeting advertising strategy might send advertising for expensive payday loans to consumers on low incomes.

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In addition, targeted advertising raises the risk of discrimination against certain consumer groups and the potential to exclude them from particular markets. And remember, all of this comes about, as pointed out by Ryan Calo in his article, by the data collected by firms from both online and in store interactions, and the use that can be made of that data through predictive analytics.

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So what can be done about this, these uses of targeted advertising? One potentially very effective response to concerns about algorithmic advertising lies in consumer data protection law. Such laws restrict the flow of data that informs predictive algorithms. In doing this, it cuts the problem off at the source - no data, no discriminatory online behavior. However, such laws are inconsistently found across nation states. Prominent examples are the

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consumer data protection regulation in the EU, and the California consumer data act. But other countries and states lack such protections. Where they do exist, data protection laws commonly require consent or contract for data collection and processing, place limits on third party sales of data, and provide consumers with the right to withdraw or reject particular data uses. By seeking consent for data collection and processing consumers are prompted to think about data use and their rights over data. So there may be an educative function here, as well as the stemming of the flow of data to digital platforms and digital advertisers. But there's a limit to the extent data protection laws can protect consumers from harms, such as those by targeted advertising and differential pricing, and primarily, this occurs because data protection laws are often premised on consumer consent. Now, it's ideal to suggest that consumers should take control over their own data uses, and to impose a robust requirement that consumers have to consent to data uses, and before information can be collected about them in their online interactions. As I've indicated, it's a useful reform that gives consumers greater control and also may educate consumers about the extent to which data is collected and repurposed. That is not sufficient to ensure consumer welfare is promoted, and that consumers' autonomy is not undermined in this new age of extensive capacity for data collection and prediction. Asking consumers to consent to data practices every time they shop or sign up for a new online service is simply likely to result in fatigue and confusion. Moreover, making decisions about privacy and data protection, behavioral biases may work against consumers making active choices that protect their interests. Consumers are limited in time, they're often overwhelmed by large amounts of information, and they exist in a state referred to by economists as bounded rationality, which means they're not always good at predicting future risks, and tend to be overconfident about their abilities to protect their own best interests. Thus, most consumer protection lawyers think there should be other policy tools providing substantive consumer protections and greater transparency around data collection and digital marketing practices. The key here is for a coordinated approach between policymakers, regulators, and consumer advocates, to continue to scrutinize the applications of new technologies in consumer markets, and to ensure that there's a variety of regulatory measures in place to ensure consumers' welfare is paramount. Consumer Protection Law also has a role to play. Blatant and extreme instances of AI being used to manipulate consumers or unduly influence them in their selection of products may be met through traditional consumer protection doctrines that provide relief when consent is in some way vitiated or undermined, so as to be meaningless in a context. For example, advertising promoting a best price guarantee for a product may be misleading, if, in fact, differential pricing is being used and other consumers obtain different or better prices. Statements about privacy protection in data collection or the security of a network device will be misleading if such protections do not exist, or are below industry standards. And regulators have recently taken action against Google for misleading conduct over privacy public protections in location tracking and in combining data from searches on non Google websites. In some instances, targeted advertising may amount to conduct that contravenes prohibitions that exist in various states on unfair commercial practices. The slide here appoints you to the trade Federal Trade Commissioner, and also a directive on unfair commercial practices in the EU. These prohibitions might apply where a data collection policy is deliberately confusing and also where, as has sometimes been suggested, consumers are identified by their online conduct as experiencing some personal or emotional crisis. And that knowledge is in use to encourage them to sign up for products that they don't need or kind of pull forward as a false way of dealing with that crisis.

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By and large, however, these types of protections although useful, also themselves may not address all of the concerns about AI in consumer markets. It's generally not misleading or unfair to send consumers advertising based on their browsing history, the time they paused on a style or type of ad, or their previous supermarket shopping. Currently, the law regards this conduct as par for the course. Now, that might be fine, we may not want

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to shut down the advertising industry, it provides amusement for many people, and also helps consumers to find new and different products. But we may, even with the protections I've mentioned, want to go a little bit further. For example, we might want strict rule-based rules in certain circumstances. And for example, we might seek to introduce rules on unfair contract terms which prevent overreaching conduct in data collection. We might want to introduce bans on targeted advertising to children and we might seek disclosure of targeted advertising or differential pricing in certain high risk contexts. We also might want to put a premium on digital media literacy education throughout society. As I mentioned before, effective response to algorithmic advertising requires regulators, advocates, consumers, lawyers and technologists to work together and to coordinate responses to ensure that AI benefits the community. It doesn't reentrench certain inequalities or harmful behaviors. Thanks. I look forward to hearing your comments on these topics.